

1 Introduction

Drought Anticipatory Action (DAA)

bnet (Bayesian network)

Standardized Precipitation Index (SPI) ranges from -4 (least rain) to 4 (most rain).

2 Notation

We will use $\mathbb{1}()$ to denote the indicator function (a.k.a. the truth function). $\mathbb{1}(S)$ equals 1 if S is true, and 0 if S is false. For example, $\mathbb{1}(x \geq 0)$ equals 0 if $x < 0$ and 1 otherwise.

In this paper, random variables will be indicated by underlining. For example, $P(\underline{x} > x)$ will denote the probability that the random variable $\underline{x}$ is greater than x .

Let

N = be the number of times (seasons) in which predicted and observed droughts were compared.

$N_{y|x}$ = number of events in which $x \in \{0, 1\}$ droughts occurred and $y \in \{0, 1\}$ droughts were predicted For example, $N_{1|0}$ is the number of events in which no drought actually occurred but it was predicted nonetheless (False Alarm)

$N_{0|0}$ = **True Negatives** (TN)

$N_{1|1}$ = **True Positives** (TP)

$N_{0|1}$ = **False Positives** (FP)

$N_{1|0}$ = **False Negatives** (FN)

$$N_{|0} = N_{0|0} + N_{1|0}, \quad N_{|1} = N_{0|1} + N_{1|1} \quad (1)$$

$$N = N_{|0} + N_{|1} \quad (2)$$

$$P(1|1) = \frac{N_{1|1}}{N_{|1}}, \quad P(1|0) = \frac{N_{1|0}}{N_{|0}}, \quad P(0|1) = \frac{N_{0|1}}{N_{|1}}, \quad P(0|0) = \frac{N_{0|0}}{N_{|0}} \quad (3)$$

3 DAA bnet

$\kappa = (\sigma, \delta)$ is the drought kind, where $\sigma \in \{\text{MAM}, \text{JJAS}\}$ ¹ is the season, and $\delta \in \{\text{Moderate}, \text{Extreme}, \text{Severe}\}$ is the drought degree.

I_κ = **Impact** (i.e., **severity**) of drought is defined as minus SPI (higher rain, higher SPI, less severe drought)

¹Letters here stand for months of the year

$P(I_\kappa)$ = likelihood, probability that a drought of severity I_κ will occur in the near future (weeks to months), based on forecasts.

μ = ensemble member index

M = number of ensemble members, 51 for *ECMWF* (European Centre for Medium-Range Weather Forecasts)

$$Risk_\kappa = E[\underline{I}_\kappa] = \frac{1}{M} \sum_{\mu=1}^M I_{\kappa,\mu} \quad (4)$$

$$= \sum_{I_\kappa} P(I_\kappa) I_\kappa \quad (5)$$

A **risk matrix** is a 4×4 matrix with X axis equal to impact (I) and Y axis equal to likelihood P , with I binned into four values and $P(I)$ binned into 4 values. The matrix elements equal to some arbitrary function $f(I, P)$ defined as risk. Usually, $f(I, P) = IP$ but not necessarily. I find risk matrices arbitrary and not very useful. The Wikipedia article on risk matrices (Ref.[1]) agrees.

$\sigma \backslash \delta$	Moderate	Extreme	Severe
MAM	-0.55	-0.98	-0.99
JJAS	-0.41	-0.98	-0.99

Table 1: SPI threshold for Karamoja, Uganda. Cells give $-I_\kappa^*$, where $\kappa = (\sigma, \delta)$

I_κ^* = impact threshold for drought (See Table 1 for examples of I_κ^* .)

$$P(\underline{I}_\kappa \geq I_\kappa^*) = \frac{1}{M} \sum_{\mu=1}^M \mathbb{1}(I_{\kappa,\mu} \geq I_\kappa^*) \quad (6)$$

P_κ^* = probability threshold for drought

C_κ = cost of drought

L_κ = loss of drought

$$P_\kappa^* = \frac{C_\kappa}{L_\kappa} \quad (7)$$

T_κ = trigger for drought

$$T_\kappa = \mathbb{1}[P(\underline{I}_\kappa \geq I_\kappa^*) > P_\kappa^*] \quad (8)$$

Let $\underline{X}_1, \underline{X}_2, \dots$ represent local road density, population density, historical catastrophic events, etc.

The TPMs² printed in blue, for the bnet Fig.1, are as follows.

²TPM=Transition Probability Matrix. This is the same thing as CPT=Conditional Probability Table

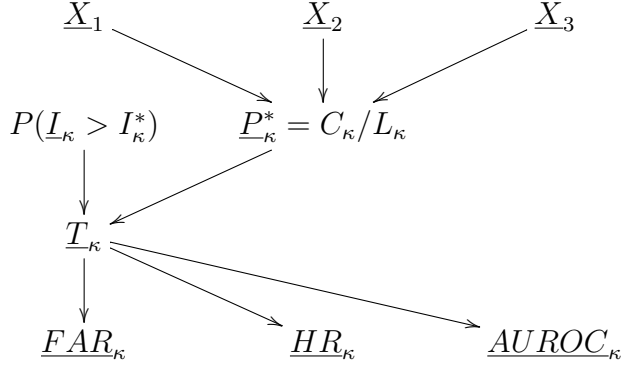


Figure 1: DAA bnet. We use one of these bnets for each κ value and for each grid cell of 0.25 degrees in latitude by 0.25 degrees in longitude. 1 degree in latitude or longitude is approximately 100 km.

$$P_{\kappa}^* = f_{\kappa}(X_1, X_2, X_3) \quad (9)$$

where $f_{\kappa}()$ is some deterministic function to be decided

$$T_{\kappa} = \mathbb{1}[P(\underline{I}_{\kappa} > I_{\kappa}^*) > P_{\kappa}^*] \quad (10)$$

$$P(Y_{\kappa}|T_{\kappa}) = \text{to be learned empirically from dataset} \quad (11)$$

where $Y_{\kappa} = FAR_{\kappa}, HR_{\kappa}, AUROC_{\kappa}$.

A Appendix: Functions of $N_{y|x}$

$$\text{Hit Rate (HR)}^3 = P(1|1) = \frac{N_{1|1}}{N_{|1}} \quad (12)$$

$$\text{False Alarm Rate (FA Rate)}^4 = P(1|0) = \frac{N_{1|0}}{N_{|0}} \quad (13)$$

$$\text{Miss Rate (MR)} = P(0|1) = \frac{N_{0|1}}{N_{|1}} \quad (14)$$

$$\text{Correct Negatives (CN)}^5 = P(0|0) = \frac{N_{0|0}}{N_{|0}} \quad (15)$$

³HR is sometimes called Precision

⁴Meteorologists also use an FA Ratio instead of FA Rate. **FA Ratio** = $\frac{N_{1|0}}{N_{1|0} + N_{1|1}} \leq 1$

⁵CN is sometimes called the **Specificity**

References

- [1] Wikipedia. Risk matrix. https://en.wikipedia.org/wiki/Risk_matrix.